

1. Summarize the dataset, including the target variable, features, and types of data

The dataset we are looking at is from CAL FIRE Historical Fire Perimeters. In this dataset, there are 22,810 records, with 21 columns. Each column would provide information about the wildfires, having data like the containment date and GIS calculated acres. There is a total of eight numerical data and 13 categorical data within the dataset. The target variable in this dataset would be GIS calculated acres, which would give us the size of the fires. We chose this as our target variable because the goal for our project is to create a model to predict the progression of California wildfires. With the GIS calculated acres as our target variable, the remaining columns would be the features such as cause, shape_area, shape_length, etc.

2. Identify and handle missing, duplicate, or inconsistent data

When looking at the dataset, we first noticed that there is missing data. Some of the columns had more missing data when compared to the others, so we decided to remove these columns as features that would affect the target variable. These columns are Complex ID, Complex Name, IRWIN ID, and Comments. When looking for duplicate data, I did `wildfire_df.duplicated().sum()`, as it would give me the total number of duplicates. The output I received from this was `np.int64(0)` which means that there are no duplicates in the dataset.

3. Calculate measures like mean, median, standard deviation, and correlations

To find measures like mean, median, and standard deviation, I did `wildfire_df.describe()` as it would give me the results I need. With this code, I have the mean, median, and standard deviation of the columns we have kept in the dataset.

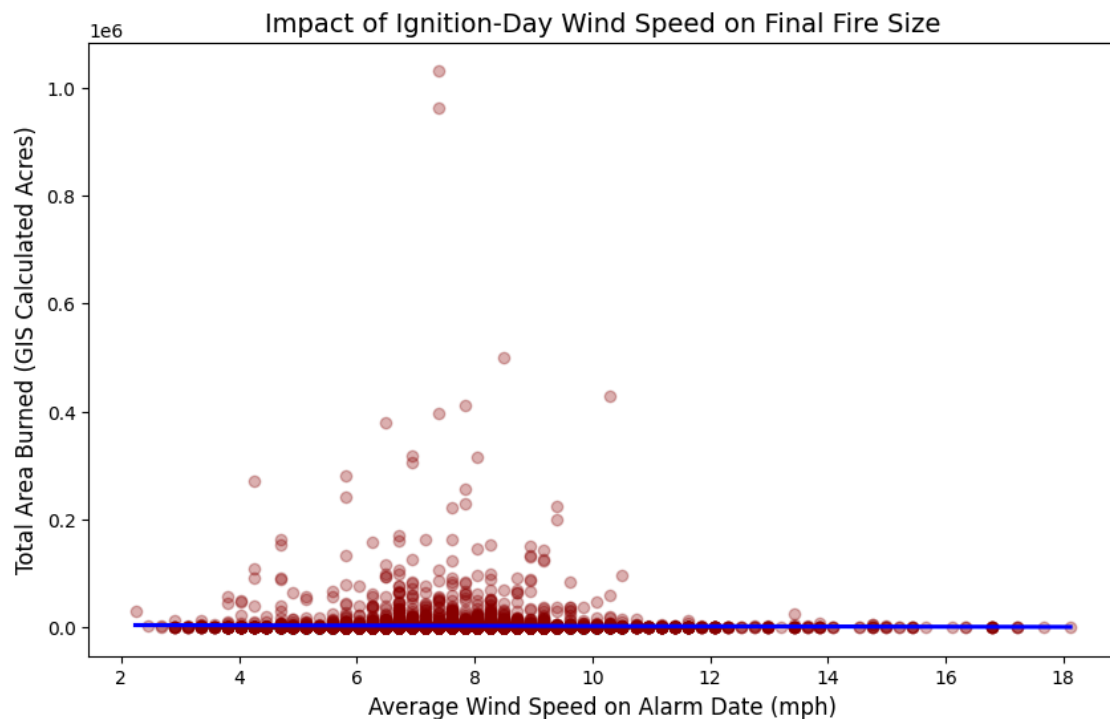
	OBJECTID	Year	Cause	Collection Method	Management Objective	GIS Calculated Acres	Shape_Area	Shape_Length
count	22810.000000	22733.000000	22810.000000	10708.000000	22539.000000	2.281000e+04	2.281000e+04	2.281000e+04
mean	11406.470627	1978.848238	9.301929	4.319014	1.012290	1.937053e+03	1.255283e+07	1.155918e+04
std	6584.873540	33.949815	5.152034	3.072826	0.110179	1.481935e+04	9.985669e+07	3.588026e+04
min	1.000000	1878.000000	1.000000	1.000000	1.000000	1.170934e-03	7.191406e+00	9.944445e+00
25%	5704.250000	1952.000000	5.000000	1.000000	1.000000	2.580515e+01	1.661103e+05	2.008347e+03
50%	11406.500000	1985.000000	10.000000	4.000000	1.000000	1.431572e+02	9.098112e+05	4.683034e+03
75%	17108.750000	2009.000000	14.000000	8.000000	1.000000	6.090518e+02	3.912756e+06	1.026799e+04
max	22811.000000	2025.000000	19.000000	8.000000	2.000000	1.032700e+06	7.116308e+09	2.010262e+06

For the correlations, I created a matrix correlation that can help identify which features would make the most difference for the target variable which is the GIS Calculated Acres. Looking at the matrix, we can see that MAX_TEMP and MIN_TEMP have 0.64 correlation, which makes sense because on warmer days, nights are usually warmer as well. There are correlations with wind speed and lagged wind meaning wind persists throughout the days which could increase fire size.

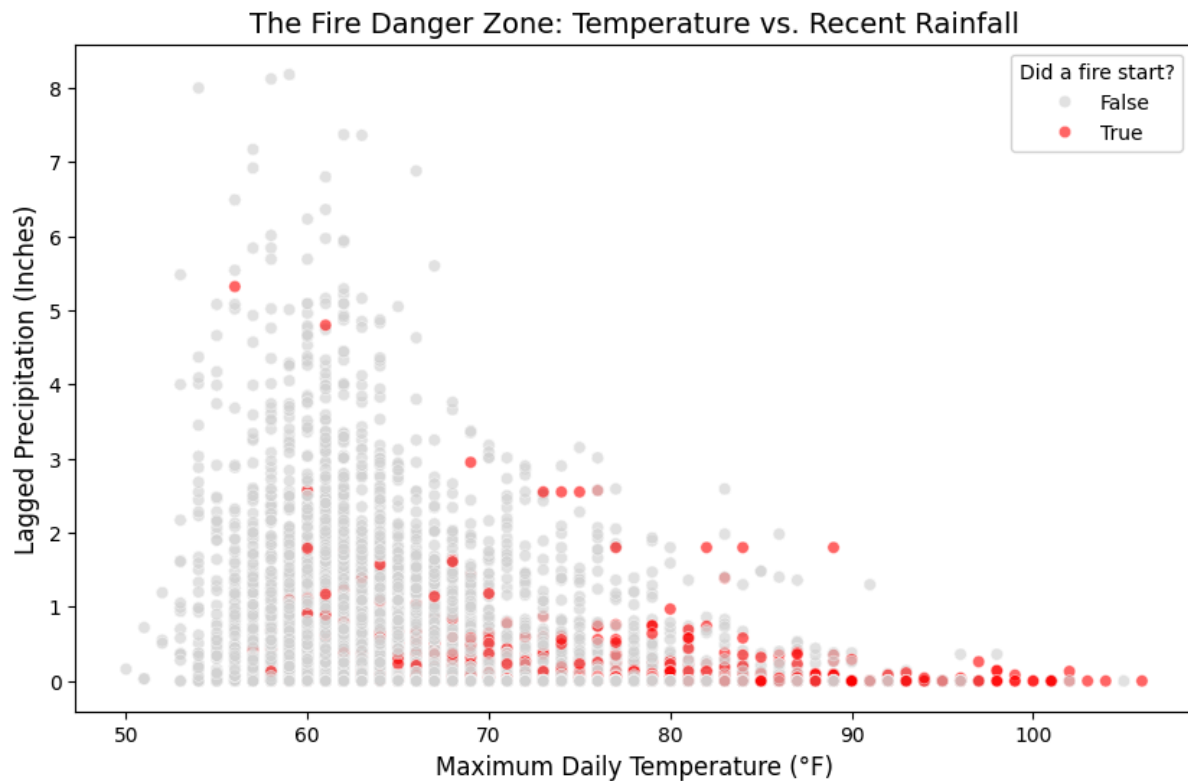


4. Create insightful visualizations (e.g., histograms, scatter plots, heatmaps)

Here is a scatter plot of wind speeds the days of fire so we can observe how much of an impact wind speed has on fire size.



The next image will depict whether lack of rain and high temperatures had any correlation with fires:



As we can see, fires increase along the x axis which is the temperature factor so it seems that temperature had the most effect on fires rather than lack of rain. There also appears to be not much correlation between lack of rain and temperature for fires starting.

With both of these visualizations, we can confidently say that temperature was the determining factor for fires given that high wind speeds and rain do not show change for number of fires.

5. Generate preliminary insights or questions to guide future analysis (hypothesis formation)

Given the following analysis of the data, we need to work on honing down whether only temperature affects fires, whether initializing them or scaling their size. Another area we can take a better look at is feature engineering. For example, is there a combination of factors that we can use which would heavily increase the accuracy of our model such as wind speed, temperature, and lag of precipitation?